Block-Insure: Hybrid Evaluation of Authority-Aware Healthcare Insurance Adjudication

***Abstract*—Healthcare insurance adjudication connects confidential evidence, external records, predictive advice, and funded decisions with distinct authority. We present Block-Insure, an Ethereum-compatible protocol evaluated using 558,211 public-secondary Medicare claims, eight controlled attacks, and executed settlement paths. Contracts bind versions, two-worker oracle agreement, four-auditor fallback, and pull payments; encrypted evidence stays off-chain and Merkle roots bind registry records. On 1,085 held-out providers, calibrated Bernoulli Naive Bayes achieved ROC-AUC 0.8797 and average precision 0.3576; logistic comparators performed better. Separate claim-routing tests raised injected-attack recall from 88.74% with rules to 100% with combined routing. All 6,160 additional detections came from provider compromise; original-cohort escalation was 7.7881%. In a 5,000-record runtime, two complete browser workflows settled through automatic processing and auditor fallback. Local tests exposed submission failures at higher concurrency and reserve inadequacy under stress. The contribution is an executable separation of advice, verification, adjudication, and settlement, evaluated against distinct evidence targets. The results support specified scenario coverage and representative execution, with broader reliability remaining unproven.**

***Index Terms*—blockchain insurance, healthcare claims, hybrid dataset, oracle consensus, risk screening, confidential evidence**

I. Introduction

A healthcare claim combines evidence, external data, decisions, and settlement. Blockchain makes transitions auditable but cannot prove that medical events occurred; oracles remain a separate trust layer [1], [2]. Models prioritize attention without establishing claim fraud. Block-Insure separates authority: contracts govern versions, roles, decisions, reservations, and balances; workers report bound observations; models advise routing; auditors resolve unsafe automation; and encrypted documents remain off-chain.

Prior healthcare-insurance systems demonstrate blockchain adjudication and rule-based controls [3]–[6]; Medicare studies evaluate imbalanced provider-risk learning [7], [8]. Public claims retain provider, beneficiary, amount, diagnosis, and time distributions but lack controlled truth for altered invoices, registry mismatch, oracle conflict, or stale versions. Deterministic mutations supply specified irregularities. This work combines both while preserving their different labels.

The central question is how to combine advice, external verification, human adjudication, and funded settlement with explicit decision authority. We ask: (RQ1) how well do aggregates rank weakly labeled held-out providers; (RQ2) where do rules and probabilistic screening complement each other on controlled attacks; (RQ3) can representative runtime paths settle; and (RQ4) what security, performance, evidence, and reserve limits remain?

We contribute (C1) a version-bound protocol combining oracle agreement, confidential evidence, auditor fallback, appeals, and pull payments; (C2) a provenance-preserving adapter for 558,211 claims with distinct provider-risk, injected-attack, and protocol outcomes; and (C3) calibrated comparisons, routing ablations, and contract, browser, evidence, reserve, and load measurements. The provider-risk study and runtime claim model are separate evaluations. Results apply to their recorded configurations; production decentralization, clinical truth, and regulatory compliance remain outside scope.

II. Background and Related Work

Mackey *et al.* prototyped multi-stakeholder blockchain claim adjudication, leaving realistic empirical validation to future work [3]. Block-HI mapped 12 fraud scenarios to blockchain controls without real-claim learning or executable end-to-end settlement evaluation [4].

B-SAHIC measured permissioned insurance performance without jointly evaluating probabilistic risk, confidential evidence, and reserve-aware settlement [5]. ChainCare evaluated an Ethereum/React DApp and gas, leaving oracle validation outside its realized trust model [9]. Islam *et al.* combined Fabric, IPFS, anonymous credentials, and fraud rules, leaving learned detection as an extension [6].

Kaur *et al.* combined explainable learning and blockchain-backed fraud records [8]; Hancock *et al.* studied imbalance and explainability in Medicare provider risk [7]. Neither jointly evaluates adversarial registry/oracle truth and complete settlement. FraudAuditor supports expert investigation of collusion, illustrating the role of human review [10].

Block-Insure evaluates their composition: distinct weak-label, injected-attack, and protocol outcomes; version-bound observations; and evidence, contract, reserve, and browser behavior within one reproducible study.

III. System and Data Architecture

A. Authority and Lifecycle

The React client authenticates by wallet signature and performs evidence cryptography. Express orchestrates authenticated requests, projections, indexing, analytics, and notifications; MongoDB is not economic authority. Two workers reconstruct registry snapshots and submit version-bound observations, treating external data as an explicit oracle boundary [1], [2]. Their agreement is application-level adjudication, separate from blockchain consensus. In the August configuration, seven Solidity modules divide state: `InsuranceManager` controls claim transitions; `PolicyEconomics` snapshots rules and

reserves coverage; `OracleCoordinator` enforces commit–reveal; `ClaimAdjudicator` owns assignment, votes, decisions, and the payout vault; `EvidenceRegistry` anchors identities and tree heads; benefits and deployment registries isolate auxiliary state.

A purchase snapshots terms and opens coverage. Submission validates ownership, incident, amount, documents, duplicate keys, and available coverage before reserving liability. Matching positive observations from both workers permit payout allocation; a negative result, conflict, or timeout becomes eligible for review. One worker alone cannot satisfy the quorum. Each review snapshots exactly four auditors: three approval votes approve, two rejection votes reject, and insufficient quorum at expiry rejects. Insufficient treasury creates a visible `FUNDING_REQUIRED` liability; after funding, claimant withdrawal produces `SETTLED`. Appeals increment claim version so old observations and decisions cannot finalize the new cycle.

B. Evidence, Registry, and Failure Containment

The threat model includes altered, duplicated, stale, or unauthorized submissions and unavailable/conflicting observations. Chain finality, signature/hash security, and correct contract execution are assumed. Registry publishers and role administrators remain trusted; both workers share one operator.

The browser encrypts with AES-GCM using claim/version/uploader/type as authenticated data [11], uploads ciphertext, and wraps the 256-bit key for proxy transformation; related PRE/blockchain work likewise separates protected health data from the ledger [12]. The backend need not learn plaintext or the key. A per-claim chain binds document order; an append-only RFC6962-style evidence tree supports inclusion and consistency proofs [13]; the chain anchors authenticated heads, not medical bytes, consistent with hybrid off-chain/on-chain storage principles [14]. Separately, oracle requests snapshot the healthcare-registry version/root, worker membership, quorum, and model identity. Missing model artifacts, disagreement, timeout, insufficient funds, unavailable storage, and stale projections fail without granting silent backend settlement authority.

C. Hybrid Dataset and Runtime

The implemented source is the compatible four-file Kaggle *Healthcare Provider Fraud Detection Analysis* Medicare benchmark [15]: provider labels, beneficiaries, inpatient claims, and outpatient claims joined only by native `Provider` and `BeneID`. It contains 558,211 claims (40,474 inpatient; 517,737 outpatient), 5,410 providers, and 138,556 beneficiaries. The public `PotentialFraud` field is a weak provider label, not claim truth; this interpretation follows provider-oriented Medicare fraud research rather than relabeling every claim [7].

For provider p and seed 20260828, the first 32 bits of $H(\texttt{split:}\|s\|p)$ assign 64%/16%/20% hash buckets to train/calibration/test. All claims of a provider remain together (Table I). Temporal features use only prior events; treatment amount ranges and 95th-percentile velocity thresholds are learned from training. Beneficiaries can cross partitions and are disclosed as a limitation.

TABLE I
HYBRID PARTITIONS AND CONTROLLED INJECTION

Quantity	Train	Calibration	Test
Claims	343,537	105,277	109,397
Providers	3,420	905	1,085
Injected attacks	40,969	12,600	54,705
Injection rate	≈12%	≈12%	50%

Independent claim-ID hashes select eight attacks: used/cancelled invoice, inflated amount, date mismatch, blacklisted provider, near duplicate, provider compromise, and noisy record. Original and modified test rows remain paired by source claim. Generative augmentation has been studied for imbalanced healthcare-fraud learning [16]; our deterministic mutations instead test specified protocol failures and do not claim to reproduce a real fraud distribution. Thus *provider risk*, *controlled attack truth*, and *on-chain protocol truth* remain separate targets.

Offline studies use all applicable claims. Runtime uses an order-independent bottom-hash sample of 4,995 real-derived training records plus five stable normal/attack presets, producing 5,000 unique invoices and a depth-13 Merkle tree. The original claim amount is reimbursement plus deductible; demonstration value is $\max(amount/10{,}000, 0.0001)$, a local scale rather than currency conversion. A manifest records hashes, counts, seed, and version `blockinsure-hybrid-v1.2.0`.

IV. METHODOLOGY

The evaluation separates weak provider labels, injected claim-level irregularities, and executed protocol outcomes (Fig. 1); none substitutes for another.

A. Provider-Risk Estimation

For provider p, nine aggregates describe claim/beneficiary volume, mean/maximum amount, inpatient/procedure/repeat ratios, mean chronic count, and activity-anomaly ratio. Training-only upper quartiles binarize these fields. Bernoulli Naive Bayes (BNB; Laplace $\alpha = 1$) estimates

$$\ell_y(\mathbf{x}) = \log \pi_y + \sum_j \left[x_j \log \theta_{jy} + (1 - x_j) \log(1 - \theta_{jy}) \right], \tag{1}$$

where π_y is class prevalence and θ_{jy} the smoothed feature probability. Softmax yields P_1; calibration providers fit a Platt sigmoid $\hat{p} = \sigma(a \operatorname{logit} P_1 + b)$ [17]. Calibration-only F_1 maximization selects $t^* = 0.35$ over $.01, \ldots, .99$; test labels never select thresholds. Comparators are majority negative, unweighted feature count, and unweighted/class-weighted logistic regression. Three leave-group-out ablations and 1,000 seeded provider bootstraps measure contribution and uncertainty [18].

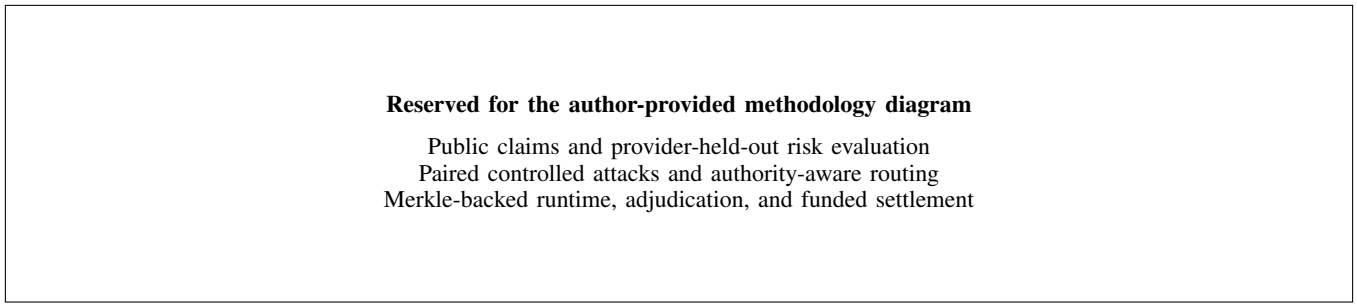


Fig. 1. Hybrid evaluation separates provider-level weak labels, paired controlled attacks, and a Merkle-backed runtime. Registry commitments establish inclusion, not clinical truth.

B. Authority-Aware Claim Routing

The *separate* protocol model uses 18 claim/registry features, not provider-risk predictions. With missed-attack, false-escalation, and review costs $(C_m, C_f, C_r) = (10, 1, 0.25)$, cost-sensitive triage [19] uses

$$t_m = \max\left(\frac{C_r + C_f}{C_m + C_f}, t_{capacity}\right) = 0.217969. \tag{2}$$

The calibration-derived capacity threshold targets a nominal 20% model-review budget; mandatory integrity routing can exceed it. Hard comparison failure H and committed/fuzzy anomaly A determine

$$\mathcal{R}(\mathbf{x}, \hat{p}) = \begin{cases} \textsc{Reject-Oracle}, & H(\mathbf{x}) = 1, \\ \textsc{Manual-Review}, & A(\mathbf{x}) = 1 \ \vee \ \hat{p} \geq t_m, \\ \textsc{Auto-Verify}, & \text{otherwise.} \end{cases} \tag{3}$$

Probability can defer to review [20], not establish authoritative rejection. Both original and paired attack cohorts contain 109,397 rows; only the latter contains the 54,705 injected positives. Model-only, rule-only, and combined routing score the same paired inputs.

Algorithm 1: Version-bound adjudication.

1: Validate policy, claim, evidence, and available coverage.
2: Snapshot claim/registry versions, root, model, and workers.
3: Workers apply integrity checks and advisory routing above.
4: Obtain version-bound commit–reveal observations.
5: Matching positive quorum permits allocation; otherwise enable review.
6: If reviewed: four seats; three approvals approve; two rejections or insufficient quorum at expiry reject.
7: On approval, allocate if funded; otherwise mark funding required.
8: Settle on claimant withdrawal. Permitted appeals restart verification under an incremented claim version.

C. Settlement, Reserve, and Evidence Experiments

For amount A, deductible rate/cap (d, D), and insurer share s (basis points), integer settlement is

$$\begin{aligned} q &= \min\big(A, \lfloor Ad/10^4 \rfloor, D\big), \\ L &= \lfloor (A-q)s/10^4 \rfloor, \qquad R = A - L, \end{aligned} \tag{4}$$

Here q is the deductible, L insurer liability, and R the patient share, subject to $reserved+settled+L \leq coverage$. Protected treasury is $U + B + \lfloor E\rho \rfloor$ for unfunded liability U, buffer B, exposure E, and fractional reserve ratio $\rho = r/10^4$ when configured as r basis points.

Reserve experiments separate frequency and severity [21]: target $K \sim Poisson(Nf)$, $X_i = \exp(\ln m + \sigma Z_i)$, with N policies, frequency f, median m, spread σ, and standard-normal Z_i. The sampler uses a normal approximation when $Nf > 30$. Capped $A_i = \min(X_i, 1)$ produces contract liability L_i and annual loss $S = \sum_{i=1}^{K} L_i$. Across M periods, capital $C = E\rho$ gives

$$P_{short} = \frac{1}{M}\sum_{j} \mathbb{1}[S_j > C], \quad \overline{D}_{short} = \mathbb{E}[S - C \mid S > C]. \tag{5}$$

The conditional mean deficit $\overline{D}_{short}$ is estimated over shortfall periods. Evidence experiments measure AES-GCM, PRE, local storage, Merkle proofs, and eight tamper/availability/access failures. Workflow tests measure execution, not clinical truth.

D. Experimental Protocol and Reproducibility

The main research snapshot is dated 28–29 August 2026, with hardening/Slither follow-up on 31 August. Figures replot those records. A separate 21 October current-source gas run uses Hardhat 2.29.1 (100M block gas limit), nine modules including hospital/bank identity registries, and two scripted settlement paths. It does not rerun the August studies; source/bytecode hashes distinguish snapshots.

The August experiments used one local x64 Windows host, Node.js 24.16.0, Solidity 0.8.26, Cancun EVM, IR, one optimizer run, Hardhat 2.28.0, and chain 31337. Oracle quorum was two, with 25-block commit/reveal windows; review required four seats with 3-approval/2-rejection thresholds. Logistic comparators used 1,200 iterations, learning rate 0.08, and L2 penalty 0.002. Reserves used 1,000 policies, 5,000 periods, seed 410034, 250 bootstrap resamples, and a 1% shortfall target. Baseline $(f, m, \sigma) = (.12, .35, .65)$; frequency stress doubles f, severity stress sets $(m, \sigma) = (.42, 1.15)$, and combined stress applies both. Claims are capped at 1 ETH, with 10% deductible capped at .02 ETH and 80% insurer share. Evidence used 30 repetitions.

Provider metrics use providers; attack metrics use paired claim rows. We report precision, recall, F_1, MCC, ROC-AUC,

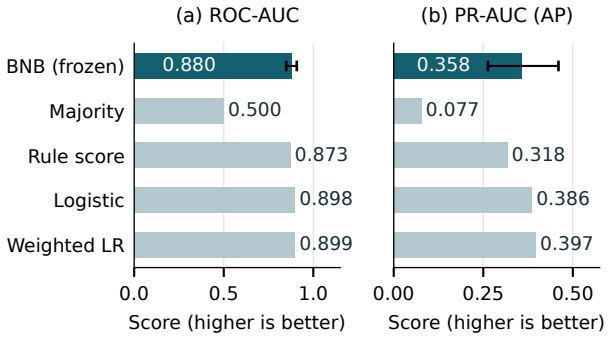


Fig. 2. Provider-held-out model comparison. Error bars are recorded 95% provider-bootstrap intervals for BNB only; none are available here for comparators. PR-AUC is average precision; positive weak-label prevalence is 7.74%.

TABLE II
PROVIDER OPERATING POINTS (WEAK LABELS)

Model	Threshold	Precision	Recall	F1
Frozen BNB	.35	.3203	.4881	.3868
Majority	.50	0	0	0
Rule score	.55	.3047	.4643	.3679
Logistic	.28	.4000	.5476	.4623
Weighted logistic	.35	.3932	.5476	.4577

average-precision PR-AUC, Brier score, and ten-bin calibration error where available. PR-AUC is emphasized under imbalance [22]. Bootstrap intervals shown for BNB are not comparator significance tests. Local latency, throughput, gas, proof size, and bytecode remain environment-specific; load batches are not replicated capacity estimates. Slither 0.11.6 [23] used identical dependency/helper-filtered scope before and after hardening. Detector observations are not confirmed vulnerabilities.

V. RESULTS

A. RQ1: Held-Out Provider Risk

Among 1,085 test providers, 84 (7.74%) carry the positive weak label. Frozen BNB produced TP=41, TN=914, FP=87, FN=43. ROC-AUC was 0.8797 (95% provider-bootstrap CI 0.8488–0.9078), PR-AUC 0.3576 (0.2631–0.4599), and F1 0.3868 (0.3011–0.4706). Majority accuracy was 92.26% despite zero recall: accuracy alone obscures screening failure.

Figure 2 and Table II show stronger logistic ranking and operating-point performance. BNB is a frozen, interpretable reference, not the best observed classifier. Removing volume/activity features caused the largest ranking loss (ΔROC=-0.1015, ΔPR=-0.1463). Removing coarse setting/clinical fields improved F1 by 0.0755, indicating encoding redundancy/noise, not clinical irrelevance.

B. RQ2: Complementary Attack Routing

On the paired cohort, combined routing produced TP=54,705, TN=50,466, FP=4,226, FN=0. Figure 3 explains the complementarity: rules detect explicit integrity failures but only 8.09% of provider-compromise attacks; the model covers provider compromise but misses inflated and noisy records. Their union reaches every tested injection. This is bounded scenario coverage, not real-fraud accuracy.

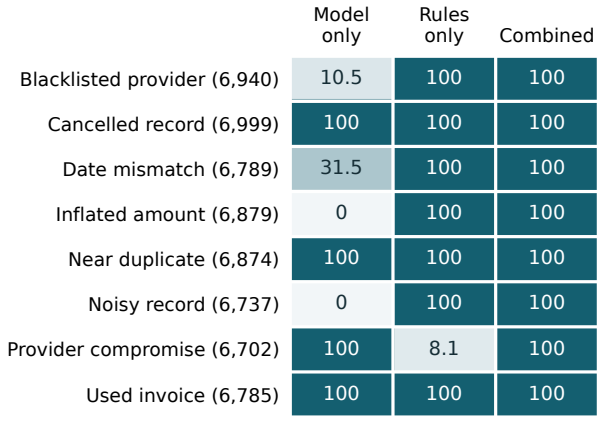


Fig. 3. Controlled-attack routing recall (%) across eight deterministic scenarios; parentheses give injected sample counts ($n = 54,705$ total). Positive routing means rejection of oracle verification or escalation, not a fraud conviction.

TABLE III
ROUTING ABLATION: ORIGINAL CONTROLS AND PAIRED ATTACKS

Strategy	Untouched routed (%)	Attack recall (%)	F1
Model only	.0037	55.25	.7117
Rules only	7.7845	88.74	.9034
Combined	7.7881	100.00	.9628

Table III's untouched rate uses all 109,397 original controls, *not* the paired cohort's negative subset. Consequently, 7.7881% is not $FP/(FP + TN)$ from the paired confusion matrix. Combined routing adds less than 0.004 percentage points of original-cohort escalation above rules. All 6,160 additional attack detections come from provider compromise; rules already cover the other seven scenarios. The gain therefore demonstrates one constructed complementarity, with generalization to other attacks untested.

C. RQ3–RQ4: Executed Workflow and Operating Boundaries

Both recorded complete browser workflows reached `SETTLED`: automatic processing and failed-oracle fallback through three auditor approvals. Table IV separates execution from operating limits. Root anchoring saved 99.62% gas against storing 500 record hashes; this is storage cost, not full-lifecycle cost. Evidence timings exclude browser and public storage/network latency.

At concurrency 50, throughput plateaued and 11/50 submissions failed (Table V); their root cause was not established. After 31 August hardening, Slither observations fell from 108 to 95: high 6 to 4, medium 28 to 22, low 65 to 62, information 5 to 5, optimization 4 to 2. Manual triage identified no unresolved exploitable defect; retained observations and 65.03% branch coverage preclude a security-proof claim. That manager build used 24,278 bytes, 298 below EIP-170 [24].

Table VI gives single local observations from 21 October; synthetic oracle inputs led both paths to funded withdrawal.

TABLE IV
AUGUST CONFIGURATION: RECORDED LOCAL MEASUREMENTS

Measurement	Result and boundary
Unit suites	Contract/backend/oracle 127/63/13; adapter 8 passed
Contract assurance	Statement/branch coverage 93.79/65.03%; sampled mutants killed 8/8
Browser workflows	Smoke/role 5/5; complete settlements 2/2 (about 84 s)
Registry/proofs	5,000 leaves, depth 13; 100/100 valid sampled proofs; mean verification .0246 ms
Registry storage gas	Root 43,851 vs. 500 hashes 11,402,190 gas
Evidence (1 MiB)	AES-GCM p95 encryption/decryption 3.432/3.341 ms
PRE key capsule	932 bytes; single-recipient p95 transformation/decryption 15.48/19.48 ms
Evidence adversarial cases	8/8 specified cases detected

TABLE V
AUGUST LOCAL SUBMISSION BATCHES (SINGLE BATCH PER LEVEL)

Concurrency	Successes/attempts	Successes/s	p95 (ms)
1	1/1	6.57	151.99
5	5/5	22.75	219.76
10	10/10	29.90	334.39
20	20/20	48.38	412.74
50	39/50	49.29	790.66

TABLE VI
OCTOBER CONFIGURATION: GAS AND PENDING L1/L2 MEASUREMENTS

Operation	Hardhat gas	L1	L2
Manager + two modules	9,435,058	–	–
Adjudicator deployment	2,451,554	–	–
Five auxiliaries (sum)	4,915,027	–	–
Deployment total (7 tx)	16,801,639	–	–
Policy purchase	731,674	–	–
Claim submission	815,312	–	–
Hospital attestation	161,288	–	–
Oracle request	448,073	–	–
Two commits + reveals (4 tx)	712,418	–	–
Open review + votes (4 tx)	871,208	–	–
Pull withdrawal	222,650	–	–

– = not measured; L1/L2 target networks pending.

Manager deployment includes coordinator/economics; auxiliaries are benefits, evidence, directory, hospital, and bank registries. Setup is excluded. Operations use the automatic path except review; rows are not an additive lifecycle. Gas is not fees; L2 needs data-cost accounting.

D. RQ4: Reserve Adequacy Under Stress

At 5% capital (Fig. 4), shortfall probability was 0.22% at baseline (95% simulation-bootstrap CI 0.10–0.34%), 28.38% under severity stress (27.34–29.43%), and 100% in high-frequency and combined runs. The smallest *tested* ratios meeting the 1% target were respectively 5%, 8%, 10%, and 15%. These grid results demonstrate sensitivity to assumptions; they establish no optimal or regulatory capital requirement.

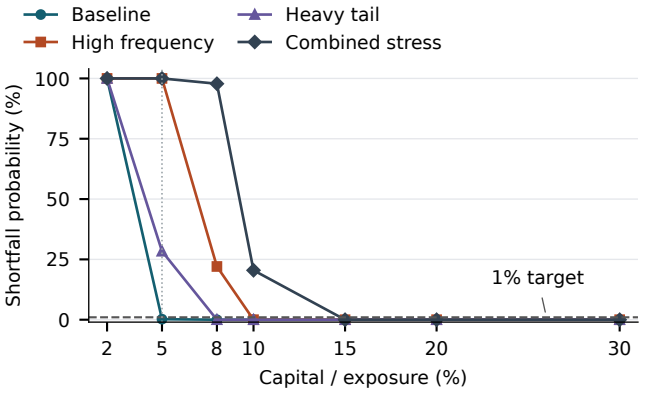


Fig. 4. Synthetic reserve stress: shortfall probability versus capital/exposure. Lines connect tested grid points, not a fitted model; dashed horizontal line marks the 1% target and dotted vertical line the configured 5% reserve. Each scenario uses 5,000 seeded periods.

VI. DISCUSSION AND LIMITATIONS

A. Interpretation

RQ1 supports weak-label prioritization: logistic comparators outperform frozen BNB. RQ2 supports complementary routing on specified injections. RQ3 demonstrates two settlement paths. RQ4 exposes reserve failures, local submission failures, bytecode pressure, and incomplete assurance. These findings do not establish unseen-fraud detection or production reliability.

The experiments answer different parts of the authority question. Provider discrimination measures advisory value; attack ablation tests specified screening behavior; executed transitions test authorization and settlement. None substitutes for the others. Probability advises review; oracle agreement, snapshotted votes, and funded withdrawal govern protocol outcomes under institutional trust. This evaluated composition is the contribution. Different labels/splits preclude cross-paper accuracy rankings.

B. Validity and Security Boundaries

Construct validity: `PotentialFraud` is inherited weak provider risk, not legal or claim-level fraud. The eight deterministic attacks exercise known fields and may be easier than adaptive behavior; 100% combined recall is therefore scenario coverage. Local settlement proves workflow execution, not clinical truth.

Internal validity: provider-disjoint splitting, training-only thresholds, calibration-only operating points, paired claims, and frozen seeds reduce leakage. Beneficiaries cross partitions, Boolean features lose information, and comparators omit boosted, graph, and sequential models. Two oracle processes share one operator; their agreement demonstrates coordination, not independent governance. Admin-published registry commitments establish inclusion, not clinical truth. Collusion and auditor unavailability remain threats: two abstentions can cause expiry rejection of a valid claim.

External validity: the public-secondary Medicare-style source may not transfer across countries, insurers, coding

systems, or time. Only 5,000 records enter runtime; all 558,211 are used offline where applicable. Hardhat automining, local MongoDB/IPFS substitutes, one Windows host, and absent CPU/RAM capture prevent production throughput claims. Evidence timings exclude browser/network storage. Poisson-lognormal reserve scenarios omit dependence, inflation, expenses, premiums, reinsurance, and regulation.

Assurance/privacy: manifests bind source/output hashes, seeds, models, compiler, bytecode, and semantic results, but independent reproduction requires the same source files. Tests, 65.03% branch coverage, eight sampled mutants, and Slither triage are neither formal verification nor a professional audit; static analysis may still miss exploitable paths or composition defects. Formal model checking has been applied to blockchain claim-processing workflows and remains complementary future assurance [25]. Browser key storage is not production custody; revocation cannot retract recovered plaintext; access metadata and availability remain observable risks. No public-testnet deployment, independent hospital oracle, penetration/privacy assessment, HIPAA/GDPR certification, or statutory reserve claim is made.

VII. Conclusion

Block-Insure implements explicit authority boundaries between statistical advice, record verification, human review, and funded settlement. Held-out provider ranking supports advisory screening; combined routing adds 6,160 detections over rules in one controlled attack family. Two browser paths settled through version-bound oracle processing or auditor fallback. Reserve stress, local submission failures, incomplete coverage, and bytecode pressure bound these results. Together, the experiments support an executable adjudication design and identify where its evidence stops.

Future work prioritizes independent reproduction, beneficiary-disjoint evaluation, adaptive attacks, independently governed data sources, public-testnet measurements, stronger assurance, and qualified actuarial calibration.

References

[1] F. Zhang, E. Cecchetti, K. Croman, A. Juels, and E. Shi, "Town crier: An authenticated data feed for smart contracts," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016, pp. 270–282.

[2] N. Goel, C. van Schreven, A. Filos-Ratsikas, and B. Faltings, "Infochain: A decentralized, trustless and transparent oracle on blockchain," in *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*, 2020, pp. 4604–4610.

[3] T. K. Mackey, K. Miyachi, D. Fung, S. Qian, and J. Short, "Combating health care fraud and abuse: Conceptualization and prototyping study of a Blockchain antifraud framework," *Journal of Medical Internet Research*, vol. 22, no. 9, p. e18623, 2020.

[4] L. Ismail and S. Zeadally, "Healthcare insurance frauds: Taxonomy and Blockchain-based detection framework (Block-HI)," *IT Professional*, vol. 23, no. 4, pp. 36–43, 2021.

[5] M. Khatun, R. A. Islam, and S. Islam, "B-SAHIC: A blockchain based secured and automated health insurance claim processing system," *Journal of Intelligent & Fuzzy Systems*, vol. 44, no. 3, pp. 4869–4890, 2023.

[6] M. M. Islam, M. Ahmed, R. Palit, M. S. Rahman, and S. Islam, "Fraud detection in privacy preserving health insurance system using Blockchain technology," *Engineering Reports*, vol. 7, no. 8, p. e70315, 2025.

[7] J. T. Hancock, R. A. Bauder, H. Wang, and T. M. Khoshgoftaar, "Explainable machine learning models for Medicare fraud detection," *Journal of Big Data*, vol. 10, p. 154, 2023.

[8] J. Kaur, G. Chawla, and A. Kumar, "Secure and explainable fraud detection in healthcare claims using blockchain-based machine learning," *Scientific Reports*, 2026.

[9] D. Verma, L. Thakare, P. Marathe, A. Agrawal, and Y. Farooqui, "ChainCare: A blockchain-based decentralized platform for transparent insurance," in *2025 IEEE 5th International Conference on ICT in Business Industry & Government (ICTBIG)*, 2025.

[10] J. Zhou, X. Wang, J. Wang, H. Ye, H. Wang, Z. Zhou, D. Han, H. Ying, J. Wu, and W. Chen, "FraudAuditor: A visual analytics approach for collusive fraud in health insurance," *IEEE Transactions on Visualization and Computer Graphics*, vol. 29, no. 6, pp. 2849–2861, 2023.

[11] M. Dworkin, "Recommendation for block cipher modes of operation: Galois/counter mode (GCM) and GMAC," National Institute of Standards and Technology, Tech. Rep. NIST Special Publication 800-38D, 2007.

[12] J. Zhang, R. Guo, Y. Shi, and W. Tang, "An anti-impersonation attack electronic health record sharing scheme based on proxy re-encryption and blockchain," *Mathematical Biosciences and Engineering*, vol. 21, no. 6, pp. 6167–6189, 2024.

[13] B. Laurie, A. Langley, and E. Kasper, "Certificate transparency," RFC Editor, Tech. Rep. RFC 6962, 2013. [Online]. Available: https://www.rfc-editor.org/rfc/rfc6962

[14] M. Pehlke, S. Fedder, C. Schmitt, E. Ndappe, N. Frochte, and M. Jansen, "Hybrid blockchain and off-chain storage for real-time data: An interval-based hashing approach," in *2025 7th International Conference on Blockchain Computing and Applications (BCCA)*, 2025, pp. 76–81.

[15] R. Anand, "Healthcare provider fraud detection analysis," Kaggle dataset, accessed: Aug. 29, 2026. [Online]. Available: https://www.kaggle.com/datasets/rohitrox/healthcare-provider-fraud-detection-analysis

[16] J. Cai, S. Wu, Y. Zhang, J. Shao, and Y. Tao, "WCGAN-GA-RF: Healthcare fraud detection via generative adversarial networks and evolutionary feature selection," *Information*, vol. 17, no. 4, p. 315, 2026.

[17] H.-T. Lin, C.-J. Lin, and R. C. Weng, "A note on Platt's probabilistic outputs for support vector machines," *Machine Learning*, vol. 68, no. 3, pp. 267–276, 2007.

[18] B. Efron, "Bootstrap methods: Another look at the jackknife," *The Annals of Statistics*, vol. 7, no. 1, pp. 1–26, 1979.

[19] C. Elkan, "The foundations of cost-sensitive learning," in *Proceedings of the Seventeenth International Joint Conference on Artificial Intelligence*, 2001, pp. 973–978.

[20] D. Madras, T. Pitassi, and R. Zemel, "Predict responsibly: Improving fairness and accuracy by learning to defer," in *Advances in Neural Information Processing Systems*, vol. 31, 2018, pp. 6147–6157.

[21] S. Gschlößl and C. Czado, "Spatial modelling of claim frequency and claim size in non-life insurance," *Scandinavian Actuarial Journal*, no. 3, pp. 202–225, 2007.

[22] T. Saito and M. Rehmsmeier, "The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets," *PLOS ONE*, vol. 10, no. 3, p. e0118432, 2015.

[23] J. Feist, G. Grieco, and A. Groce, "Slither: A static analysis framework for smart contracts," in *2019 IEEE/ACM 2nd International Workshop on Emerging Trends in Software Engineering for Blockchain (WETSEB)*, 2019, pp. 8–15.

[24] V. Buterin, "EIP-170: Contract code size limit," Ethereum Improvement Proposals, 2016. [Online]. Available: https://eips.ethereum.org/EIPS/eip-170

[25] R. L. Neupane, E. Bonnah, B. Bhusal, K. Neupane, K. A. Hoque, and P. Calyam, "Formal verification for blockchain-based insurance claims processing," in *2024 IEEE/IFIP Network Operations and Management Symposium (NOMS)*, 2024.